Chapter 02 Homework

Part A multiple choice (select the one that is best in each case. 1point/question)

1. Which of the following nonmetals will form an ionic compound with Sc^{3+} that has a 1:1 ratio of cations to anions ?
A) Ne B) F C) O D) N E) H

2. Which of the following factors determines the size of an atom?
A) the volume of the nucleus
B) the volume of space occupied by the electrons of the atom
C) the volume of a single electron, multiplied by the number of electrons in the atom
D) The total nuclear charge
E) The total mass of the electrons surrounding the nucleus.

3. The gold foil experiment performed in Rutherford's lab ________.
A) proved the law of multiple proportions
B) utilized the deflection of beta particles by gold foil
C) confirmed the plum-pudding model of the atom
D) led to the discovery of the atomic nucleus
E) was the basis for Thomson's model of the atom

4. Which of the following statements is *false*?
A) the nucleus has most of the mass .
B) every atom of a given element has the same number of protons.
C) the number of protons in an atom equals the number of neutrons in the atom.
D) electrons comprise most of the volume of an atom.
E) the number of electrons in an atom equals the number of protons in the atom.

5. Predict the molecular compounds of the following species: $HClO_4$, CH_3OCH_3, $Mg(NO_3)_2$, H_2S, $TiCl_4$, K_2O_2, PCl_5, $P(OH)_3$.
A) $HClO_4$, CH_3OCH_3, $Mg(NO_3)_2$ B) H_2S, $TiCl_4$, K_2O_2
C) CH_3OCH_3, $Mg(NO_3)_2$, H_2S D) K_2O_2, PCl_5, $P(OH)_3$
E) $HClO_4$, CH_3OCH_3, H_2S, PCl_5, $P(OH)_3$

6. How many protons, neutrons, and electrons are in an atom of $^{197}Au^{+}$?
A) 79, 79,118 B) 79, 118, 78 C) 118, 79, 79 D) 79, 197, 78 E) 197, 79, 79

7. Of the three types of radioactivity characterized by Rutherford, which type does not become deflected by a magnetic field?
A) β-rays
B) α-rays and β-rays
C) α-rays
D) γ-rays
E) α-rays, β-rays, and γ-rays

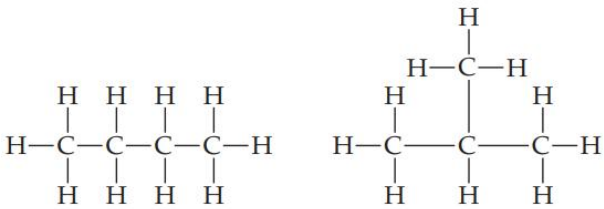
8. Which one of the following is not one of the postulates of Dalton's atomic theory?
A) Atoms are composed of protons, neutrons, and electrons.
B) All atoms of a given element are identical; the atoms of different elements are different and have different properties.
C) Atoms of an element are not changed into different types of atoms by chemical reactions: atoms are neither created nor destroyed in chemical reactions.
D) Compounds are formed when atoms of more than one element combine; a given compound always has the same relative number and kind of atoms.
E) Each element is composed of extremely small particles called atoms.

9. In the Rutherford nuclear-atom model, ________.
A) the three principal subatomic particles (protons, neutrons, and electrons) all have essentially the same mass
B) mass is spread essentially uniformly throughout the atom
C) the heavy subatomic particles, protons and neutrons, reside in the nucleus
D) the light subatomic particles, protons and neutrons, reside in the nucleus
E) the three principal subatomic particles (protons, neutrons, and electrons) all have essentially the same mass and mass is spread essentially uniformly throughout the atom

10. In the periodic table, the elements are arranged in ________.
A) order of increasing neutron content
B) order of increasing metallic properties
C) order of increasing atomic number
D) alphabetical order
E) reverse alphabetical order

11. The charge on an electron was determined in the ________.
A) cathode ray tube, by J. J. Thomson
B) Rutherford gold foil experiment
C) Millikan oil drop experiment
D) Dalton atomic theory
E) atomic theory of matter

12. The structural formulas of the compounds n-butane and isobutane are shown below.Which formula(s) allow you determine these are different compounds?



n-butane isobutane

A) empirical formula
B) structural formula
C) molecular formula
D) both of the above

13. Which pair of substances could be used to illustrate the law of multiple proportions?
A) SO_2, H_2SO_4
B) CO, CO_2
C) H_2O, O_2
D) CH_4, $C_6H_{12}O_6$
E) $NaCl$, KCl

Chapter 02 Homework

Part B: Short Answer -- Write legibly and show all work for all steps in the problem. (12 points)

14. Give the name or chemical formula, as appropriate, for each of the following substances: (a) $Fe_2(CO_3)_3$, (b) sodium hypobromite, (c) HBr, (d) sulfurous acid, (e) SF_6

15. From the following list of elements—Ar, Ga, Ca, Br, Ge—pick the one that best fits each description. Use each element only once: **(a)** an element that resembles aluminum, **(b)** an alkaline earth metal, **(c)** a noble gas, **(d)** a halogen, **(e)** a metalloid.

16. Iron has three major isotopes: ^{54}Fe (atomic mass = 53.9396 u; abundance 5.85%), ^{56}Fe (atomic mass =55.9349 u; abundance 91.8%), and ^{57}Fe (atomic mass = 56.9354 u; abundance 2.35 %). Calculate the atomic weight (average atomic mass) of iron.